Advanced programming Assignment

ROLL NO: CSB24065

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Question:

Design a student system in Python with:

Address class (street, city, zipCode)

Student class with name, age, Address, and course list

Store age as a protected attribute and control it using @property

Methods: add_course() and display()

Extend it with:

ScholarshipStudent (add scholarshipAmount and override display())

Your implementation should clearly show:

Composition (Student HAS-A Address) Proper data validation using @property (age must be valid)

Inheritance and overriding (use super() in display)

Understanding of mutable behavior (course list updates persist)

Answer:

```
class Address:
    def __init__(self, street, city, zipCode):
        self.street = street
        self.city = city
        self.zipCode = zipCode

    def display(self):
        return f"{self.street}, {self.city}, {self.zipCode}"

class Student:
```

```
    def __init__(self, name, age, address, courses=None):
        self.name = name
        self._age = 0
        self.age = age
        self.address = address
        self.courses = courses if courses is not None else []

    @property
    def age(self):
        return self._age

    @age.setter
    def age(self, value):
        if value <= 0 or value > 120:
            raise ValueError("Invalid age")
        self._age = value

    def add_course(self, course):
        self.courses.append(course)

    def display(self):
        print("Name:", self.name)
        print("Age:", self.age)
        print("Address:", self.address.display())
        print("Courses:", self.courses)

class ScholarshipStudent(Student):
    def __init__(self, name, age, address, courses, scholarshipAmount):
        super().__init__(name, age, address, courses)
        self.scholarshipAmount = scholarshipAmount
```

```
    def display(self):
        super().display()
        print("Scholarship Amount:", self.scholarshipAmount)

addr1 = Address("MG Road", "Tezpur", "784001")

s1 = Student("Gourisankar", 20, addr1)
s1.add_course("Python")
s1.add_course("Data Structures")
s1.display()

print()

s2 = ScholarshipStudent("Rahul", 21, addr1, ["Java"], 5000)
s2.add_course("Machine Learning")
s2.display()
```

Screenshots of the output: